

EXP 9: Compare Linear and Polynomial Regression using Python.

AIM

To compare the performance of Linear Regression and Polynomial Regression using Python.

SOFTWARE USED

1. Google Colab
2. Python 3.x
3. NumPy
4. Matplotlib
5. Scikit-learn
- 6.

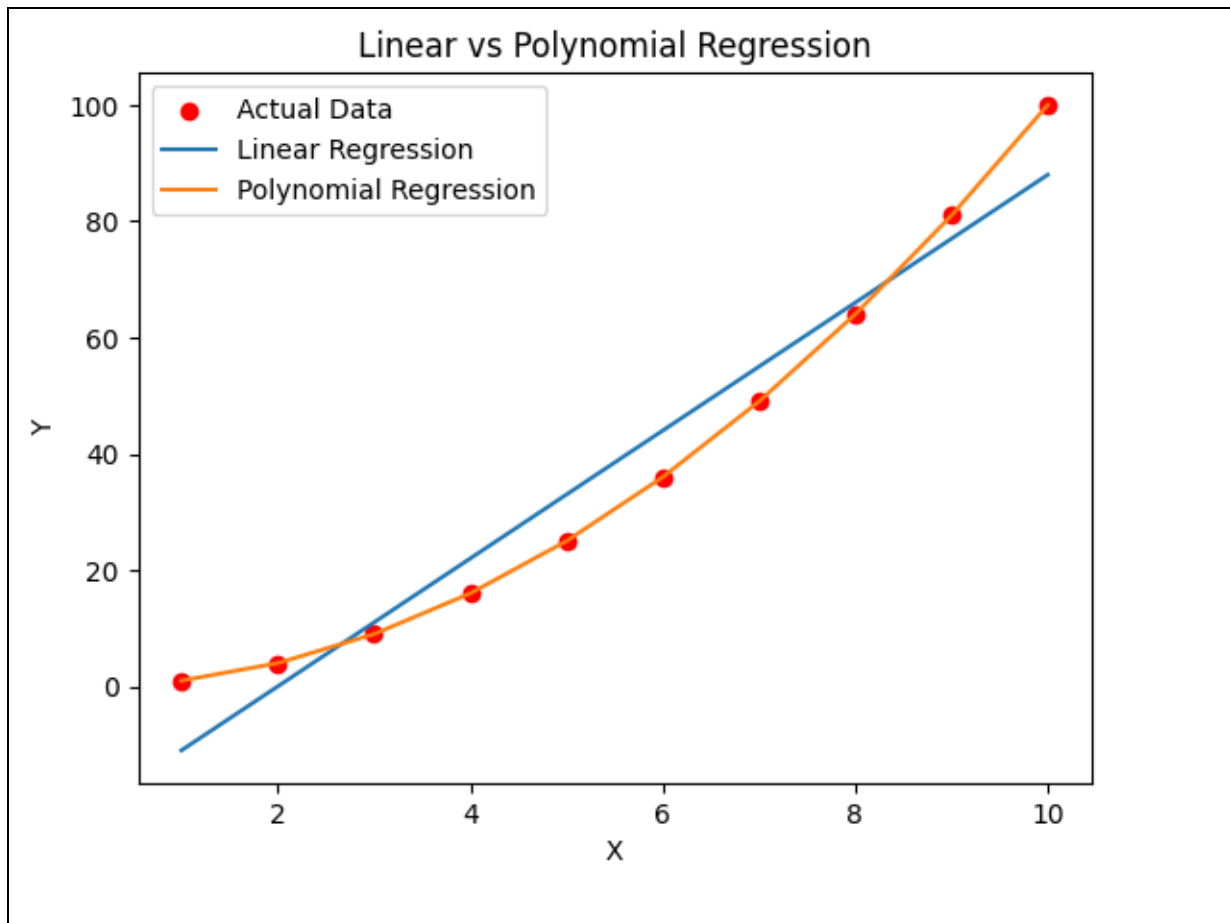
PROCEDURE

1. Import the required libraries.
2. Create a sample dataset.
3. Train a Linear Regression model.
4. Transform the data into polynomial features.
5. Train a Polynomial Regression model.
6. Compare the predictions of both models.
7. Plot the results.

GOOGLE COLAB CODE

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
# Sample Dataset
X = np.array([[1],[2],[3],[4],[5],[6],[7],[8],[9],[10]])
y = np.array([1,4,9,16,25,36,49,64,81,100])
# Linear Regression
linear = LinearRegression()
linear.fit(X, y)
# Polynomial Regression
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)
poly_model = LinearRegression()
poly_model.fit(X_poly, y)
# Predictions
linear_pred = linear.predict(X)
poly_pred = poly_model.predict(X_poly)
# Plot
plt.scatter(X, y, color='red', label='Actual Data')
plt.plot(X, linear_pred, label='Linear Regression')
plt.plot(X, poly_pred, label='Polynomial Regression')
plt.title("Linear vs Polynomial Regression")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.show()
```

OUTPUT



A graph is displayed comparing:

- Linear Regression (straight line)
- Polynomial Regression (curve fitting the data)

RESULT

Thus, Linear Regression and Polynomial Regression were implemented and compared. Polynomial Regression provided a better fit for the non-linear dataset.